DRILS Revisited: On the Combination of Perturbation with Graybox Optimization Techniques

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ABSTRACT

Designed for graybox optimization problems, the state-of-the-art DRILS algorithm (Deterministic Recombination and Iterated Local Search) follows the framework of a hybrid iterated local search by combining the efficient identification of improving moves and the fast recombination of local optima. The DRILS algorithm uses a perturbation mechanism in order to feed the graybox crossover with promising local optima. The perturbation is a key element to avoid that the search gets trapped. In this paper, we revisit the DRILS algorithm by focusing on two main questions: (i) how the perturbation is performed, and (ii) how strong it should be. We propose two alternative designs of the perturbation within the framework of DRILS. The so-obtained algorithms are proved to provide substantial improvements. This is demonstrated based on extensive experiments using a diverse set of NKQ-landscapes, with different degrees of ruggedness, as well as, different dimensions ranging from relatively small to very large. Besides, we provide a comprehensive analysis on the impact of the proposed mechanisms allowing us to highlight the guiding principles for an accurate design and configuration of the perturbation as a function of landscape characteristics.

CCS CONCEPTS

• **Theory of computation** → *Discrete optimization.*

KEYWORDS

Graybox optimization, NK-landscapes, perturbation

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1 INTRODUCTION

Context. Evolutionary *graybox optimization* [6, 20] refers to techniques that take benefits from the knowledge that is available a priori about the optimization problem. This is the case of a number of fundamental optimization problems, e.g., TSP, QAP, NK/MK landscapes, Max-kSAT, etc. The structural information underlying the formulation of such problems is extremely important in order to develop accelerated neighborhood-based move operators [4, 19] and genetic recombination [2, 3, 17]. The success of these operators comes from their ability to perform an informed, *not* blind, exploration of a large set of candidate solutions in a very efficient manner. In this paper, we focus on the so-called *DRILS (Deterministic Recombination and Iterated Local Search)* algorithm [5], which is considered as a state-of-the-art framework for the integration and the validation of informed evolutionary graybox operators.

More specifically, we are interested in revisiting the general design of the DRILS framework for solving k-bounded functions [20, 21]. The DRILS algorithm was in fact successfully applied to tackle the well-established NK-landscapes with hundred thousands up to one million variables [5]. *NK-landscapes* [9, 10] are pseudo-boolean functions of dimension n (i.e. assigning real values to bit-strings of size n) which are written as a linear combination of n sub-functions. Each sub-function accepts a restricted number of variables, namely, $k = K + 1$ variables, with K a constant parameter. Hence, the degree of non-linearity of the function is mainly impacted by the k-value.

In a graybox optimization setting, the interaction scheme among variables is assumed to be known. This knowledge is modeled by the so-called *variable interaction graph (VIG)* [20]. The strength of DRILS is mainly due to its ability to smartly combine the VIG-informed operators, by following a hybrid iterated local search like scheme. Roughly speaking, the DRILS framework has two main components: (i) one intensification component which is the by-product of combining graybox local search and recombination operators, and (ii) one diversification component which consists in performing a perturbation. In this paper, we are specifically interested in revisiting the design of the diversification components which, to the best of our knowledge, was addressed in very few studies.

Related work and motivation. DRILS combines two powerful ideas. The first one, namely, *efficient hamming ball hill-climber* [4], is able to identify improving moves in constant time under some conditions. The second one, namely *Partition crossover* [17], is able to find in linear time the best offspring among the eventually huge number of possible recombinations of non-identical gene-components from two given parents. In DRILS, these two parents are actually two *local optima* found by running the efficient hamming ball hill-climber. In more details, DRILS maintains a first parent on which a perturbation is performed. Then, by using the efficient hill-climber, a new local optima constituting the next parent is constructed starting from the so-obtained perturbed solution. After recombination, the current first parent is updated and the whole process is iterated. In [5], DRILS was found to outperform other graybox algorithms, namely, Hierarchical Recombinative Local Search (HIRELS) proposed in the

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same paper, and the Gray-Box Parameterless Population Pyramid algorithm (GB-P3) [6]. Since then, DRILS was accepted to be the state-of-the-art, and the latest developments have focused on integrating new graybox operators [2, 3] within the DRILS framework. This is mostly guided by allowing the partition crossover to intensify the search; hence, eventually putting aside another important concept in DRILS which is the design of the perturbation. This is precisely the main motivation of our work.

Taking a look at the lessons learnt from the past [7, 11, 15], the basic working principle underlying an *iterated local search* (ILS) algorithm is to perform a biased walk in the space of local optima. Apart from the local search component, the perturbation allowing to jump to new basins of attraction, and the acceptance criteria allowing to decide which local optima to consider in the next iteration, are critically important. All these components are strongly inter-dependent and their smart combination is the key in order for an ILS algorithm to perform at its best for a given problem. In particular, the choice of the perturbation strength is believed to relate to the structure of the underlying landscape, e.g., [12, 14] to cite a few. With respect to k-bounded functions, the experiments conducted initially in [4] show that the value of the best performing perturbation strength in DRILS depends on k. Besides, the perturbation impacts the number of recombination components identified by the crossover, which at the same time impacts the performance of the crossover, and subsequently the whole search process. Quoting the comments in [4], it is believed that "there is an inverse relationship between the optimal perturbation factor and [DRILS] performance".

Contribution overview. Our work is based on the general requirement of having a good balance between intensification and diversification in order for any evolutionary algorithm to achieve optimal performances. Since the graybox operators in DRILS are extremely effective in finding improving solutions, one should make sure that they are accurately complemented by the perturbation, which is the rule-of-thumb when designing ILS. Besides, when facing very rugged landscapes, diversifying the search is crucial, and the perturbation is critical in order to jump over diversified high-quality local optima, eventually helping to approach the global optimum. In this paper, we propose to revisit the design of DRILS in light of the perturbation component in an attempt to address two main issues: (i) how the perturbation is performed, and (ii) how strong it should be. Our contribution can then be summarized as follows:

- We propose two simple alternative designs for setting up the perturbation in DRILS. The first alternative is a simple modification allowing to couple more tightly the perturbation with the acceptance criteria in an attempt to escape local optima as soon as the graybox operators are not able to improve. The second alternative consists in disabling the crossover occasionally, hence allowing the local search to find fresh local optima that are eventually more recombination promising.
- We show that the so-obtained DRILS variants provide substantial improvements over the initially employed perturbation mechanism. We emphasize that our goal is not to horse-race against DRILS, but to gain a better understanding of the importance of designing perturbation. As such, our results show that the perturbation can play a major role in enhancing existing graybox optimization techniques.
- Our results are accompanied with a comprehensive empirical analysis of DRILS performance profile using a diverse set of random NKQ-landscapes, with different degrees of ruggedness, as well as, different dimensions ranging from relatively small to very large. In particular, we study the impact of the different parameters introduced by the proposed designs of perturbation, and highlight the guiding principles for an accurate configuration of the perturbation as a function of the landscape characteristics.

Outline. In Section 2, we recall the necessary background. In Section 3, we recall the DRILS framework and conduct a preliminary analysis of its initial perturbation mechanism. In Section 4, we revisit the design of perturbation in DRILS, and report our empirical findings. In Section 5, we conclude the paper.

2 BACKGROUND

2.1 k-bounded functions and NKQ-landscapes

We consider the class of k-bounded pseudo-boolean (unconstrained) optimization problems [20]. A pseudo-boolean objective function f (to be maximized) of n binary variables assigns a real value to every solution, i.e., $f : \{0,1\}^n \mapsto \mathbb{R}$. The function is said to be k-bounded if it can be written as the sum of m sub-functions $f^{(\ell)}$, each one depending on at most k variables:

$$f(x) = \sum_{\ell=1}^{m} f^{(\ell)}(x_{i_{\ell,1}}, x_{i_{\ell,2}}, \ldots, x_{i_{\ell,j\leq k}}) \quad (1)$$

where $i_{\ell,j}$ is the index of the jth variable ($j \leq k$) of $f^{(\ell)}$. A gray-box optimizer is given access to Eq.(1).

Following previous studies [2–4], we will work with a representative class of k-bounded functions, namely NKQ (Quantized NK) landscapes, with one subfunction per variable ($n = m = N$). For every $i \in \{1, \ldots, n\}$, the subfunction $f^{(i)}$ depends on variable x_i and other $K = k-1$ variables. The codomain of each sub-function is the set $\{0, 1, \cdots, Q-1\}$, where Q is a positive integer. The parameter K defines the number of epistatic interactions. Different models can be used to initialize the subfunctions [8, 13]. For our experimentations, we consider a random model. More precisely, the K epistatic interactions are selected at random, and the sub-functions values are generated at random.

2.2 Graybox tools and evolutionary operators

Graybox optimization techniques for k-bounded functions are based on a graph to represent the knowledge about the interactions among variables. This graph is the cornerstone for the development of the efficient operators used within the DRILS framework.

2.2.1 Variable interaction graph (VIG). A VIG [20] is a graph where the set of nodes corresponds to the binary variables, and there exists an edge between two nodes if there exists a *non-linear interaction* between the corresponding variables. Non-linear interactions can be captured either by assuming that every pair of variables appearing in the same sub-function have a nonlinear interaction, or by applying a Fourier transform [16] and then looking at the non-zero coefficients associated to a term with every two variables.

2.2.2 Efficient Hamming Ball Hill-Climber. A Hill-Climber consists in moving from one solution to a new improving one following a neighborhood relation. We consider a neighborhood based on the Hamming distance, i.e., two bit-string solutions are at Hamming distance r if they differ by r bits. Even when neglecting the cost of sub-function evaluation, searching for an improving neighbor by a naive (blackbox) enumeration approach can be prohibitive even for $r = 1$. The secret for a fast identification of an improving neighbor in the r-Hamming ball [1, 4, 19] is based on the observation that a bit-flip impacts a restricted number of sub-functions. Assuming that each variable appears in $O(c)$ sub-functions, this can be performed in $O((3kc)^r)$ time complexity which is independent of the problem dimension n (see Theorem 1 in [4]). This leads to a time-efficient first-ascent hill-climber which serves as the local search component of the DRILS algorithm. Following the existing literature [2–4], we consider the hamming distance 1, and we denote by EFFICIENTHAMMINGBALLHC the so-obtained local search.

2.2.3 Deterministic recombination. The second core component of DRILS is the *Partition crossover* (PX) which is a greedy deterministic recombination operator, studied in [18] for TSP, and leveraged for k-bounded functions [17]. Given two parents p_1 and p_2, PX follows a gene transmission principle by identifying the variables having the same assignments in the two parents. These variables are removed from the VIG, hence partitioning the graph (and the underlying variables) into q (non-interacting) connected components, constituting the recombination components. The variables that appear *together* in the same component are inherited *together* from either p_1 or p_2. This partitioning process infers a linearly separable function. Thus, among the 2^q offsprings that can be derived, the best one can be computed in linear time. When q is large, partition crossover is extremely efficient to find improving solutions (if any) in a huge proportion of the search space, namely $\Theta(2^q)$. Besides, when the parents are local optima, PX allows to jump through the local optima space, eventually acting as a tunneling algorithm.

Putting the pieces together, we describe in the next section the original design of DRILS. We also conduct a preliminary analysis of DRILS preparing for the main contribution of the paper.

Algorithm 1: DRILS (Deterministic Recombination and Iterated Local Search) [5]

```
1  x ← initial solution;
2  p1 ← EfficientHammingBallHC(x);
3  while not stopping condition do
4      x ← Perturbation(p1, α);
5      p2 ← EfficientHammingBallHC(x);
6      p ← PartitionCrossover(p1, p2);
7      if p1 = p or p2 = p then
8          p1 ← p2;
9      else
10         p1 ← EfficientHammingBallHC(p);
```

3 A RETROSPECTIVE OUTLOOK OVER DRILS

3.1 Discussion on the initial design of DRILS

3.1.1 Description of DRILS. The DRILS algorithm is depicted in Algorithm 1. DRILS mixes efficient hamming ball hill-climber with recombination following the basic scheme of an iterated local search. More specifically, DRILS recombines consecutive local optima. It starts with a first parent p_1 initialized with a locally optimum solution computed using the efficient hamming ball hill-climber. In order to feed the crossover, DRILS perturbs the first parent p_1 by randomly flipping $\alpha \cdot n$ bits. The (meta-)parameter α is called the strength of the perturbation. This perturbation constitutes a soft-restart allowing to have a starting solution x (line 4), from which a second locally optimum parent p_2 is computed using the graybox hill-climber (line 5). The partition crossover is then applied in order to identify the best possible child, denoted p in line 6.

Two situations can then happen. In the first case (line 7), the child p is similar to either p_1 or p_2, which means that the crossover was not successful in finding an improving child. Then, the current first parent p_1 is replaced by p_2, and the search is resumed. Otherwise (line 9), this means that the child p improves over its parents, and the efficient hill-climber is applied again so to have a new local optimum p_1 ready for the next recombination. The whole process is repeated until a termination condition is satisfied.

3.1.2 Discussion. We can notice two important aspects in the design of DRILS. Firstly, DRILS always perturbs *and* recombines the most recently discovered local optima. This holds independently of whether the child solution is improving or not. Taking a more measured look, this relates to the choice of the acceptance criteria in standard iterated local search [7, 11, 15]. Secondly, the perturbation plays two roles. On the one hand, it generates the starting solution to construct the second parent. For the graybox crossover to be effective, the two parents should be two local optima being at a relatively close hamming distance one from the other. Hence, the perturbation strength α should typically be small to avoid ending up with too different parents. However, since a local search is applied to obtain the second parent, the perturbation alone cannot fully control how far are the two parents. Thus, the perturbation strength α should be enough strong to allow the local search to generate a reasonably close, but different, second locally optimum parent. On the other hand, since the perturbation strength highly influences the newly generated parent, it consequently influences the child p, and hence the choice of the next current first parent to recombine in line 7. Said differently, the influence of the perturbation goes beyond one single iteration of DRILS, and can eventually help the crossover generating high-quality parents in subsequent iterations.

In order to better understand how relevant are these design choices, we propose to conduct a preliminary, yet detailed, analysis of DRILS. This preliminary analysis shall allow us to introduce the new design options we are proposing in this paper.

3.2 Experimental analysis of DRILS

3.2.1 Baseline design choices. For a better understanding of DRILS guiding principles, we consider to contrast the relative performance of DRILS with two different baseline variants: (i) *not* selecting the most recently discovered local optimum as a first parent, and (ii) *systematically* disabling the graybox crossover.

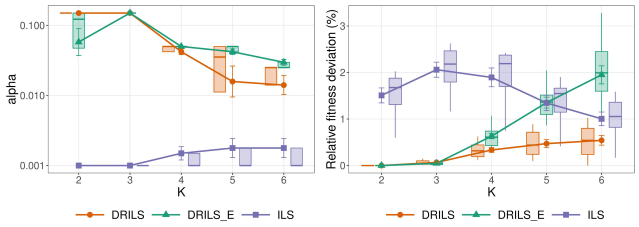


Figure 1: Left (respectively, Right): the best perturbation strength α (respectively, the corresponding relative fitness deviation) as a function of k. Notice that the *log* scale for α.

The first option consists in removing the conditional instruction in lines 7, 8 and 9, and keeping line 10 in Algorithm 1. The so-obtained algorithm is called DRILS$_{\text{elite}}$. In fact, the child p can only be better or equal to its parents, and DRILS$_{\text{elite}}$ always keeps the best solution found so-far as the current first parent. This can be viewed as an elitist design of the acceptance criterion, which is to contrast with the initial design of DRILS. The second option is a purely local search based algorithm performing perturbation followed by the hamming ball hill-climber, and accepting the newly generated local optimum. This consists in removing the whole instruction block *from* line 6 *to* 10, and replacing p_2 in line 5 by p_1. We denote by ILS the so-obtained algorithm variant. Notice that this ILS variant has a non-elitist design of the acceptance criterion.

3.2.2 Experimental set-up. Following previous work [2–4], we consider random NKQ landscapes with $Q = 64$. The degree of non-linearity is set in the range $k = K + 1 \in \{2, 3, 4, 5, 6\}$. The problem dimension is set in a range $n \in \{10^3, 10^4, 10^5, 10^6\}$ spanning relatively small and very large instances of one million variables. We experiment a perturbation strength varying in the range $\alpha = r \cdot 10^{-2}$; $r \in \{0.1, 0.25, 0.5, 1, 2.5, 5, 10, 15\}$. Each algorithm is executed for 10 independent runs. The termination condition is set to 15 minutes. This represents $3_{(\text{algos})} \times 5_{(k)} \times 4_{(n)} \times 8_{(\alpha)} = 480$ configurations requiring 50 CPU days to run.

All algorithms are implemented in `C++`. We took a special care having an extremely efficient implementation of DRILS, e.g., the scores required for the hill-climber are copied whenever possible after the perturbation and crossover. Each configuration run is executed on a `Intel Xeon Gold 5220` CPU with `96 GiB RAM`.

3.2.3 Analysis. Due to space limitations, we only provide a global overview of DRILS performance in this preliminary analysis. For every NKQ landscape and every algorithm, we compute the value of α providing the *best* fitness value in average. In Fig. 1, we report the average (over n) of the (per-instance) best setting of the perturbation strength α. Besides, we report the performance of each algorithm configured with the (per-instance) best setting of α. For the sake of normalization, the performance is measured as the *percentage fitness deviation* of the best solution, say s, found by an algorithm during a run, from the best solution, say $\tilde{s}^\star$, found by any run for a given instance, i.e., $100 \cdot (f(\tilde{s}^\star) - f(s))/f(\tilde{s}^\star)$. The lower the better.

Without surprise, DRILS is the best performing algorithm. However, we can extract a number of interesting observations informing about the impact of the design choices. Firstly, the best setting of the perturbation strength α depends both on the degree of non-linearity and the considered algorithm. In the case of DRILS, and in line with [4], the general trend of the best value of α decreases as k increases. This general tendency is the same for DRILS$_{\text{elite}}$, but with the α-values being slightly larger as the difficulty of the landscape increases. Notice that DRILS$_{\text{elite}}$ has a comparable performance to DRILS for small k, while it is the worst performing algorithm for a large k. This can be attributed to the fact that the perturbation alone in DRILS$_{\text{elite}}$ is not able to properly escape from local optima for the hardest instances. Interestingly, the general tendency observed for DRILS and DRILS$_{\text{elite}}$ is different when analyzing the best α-values found for ILS. In this case, the range of α is significantly smaller. As the difficulty of the landscape increases with k, the relative difference between DRILS and ILS decreases, with ILS requiring larger values of α. For the largest $k = 6$ representing the most rugged/difficult landscapes, ILS is even better than DRILS$_{\text{elite}}$. From these observations, we can draw two main conclusions:

(1) The choice of the first parent in DRILS is important, and should be accompanied with an accurate choice of the perturbation strength when preparing for the second parent (as supported by the relative behavior of DRILS and DRILS$_{\text{elite}}$).
(2) A pure (non-elitist) iterated local search with a carefully chosen strength can help to escape from local optima (as supported by the relative behavior of DRILS$_{\text{elite}}$ and ILS).

The previous two observations constitute the rational behind our contribution. They suggest that there could be other alternative choices enhancing the initial design of DRILS. We shall demonstrate this claim by proposing and studying two different design options.

4 DRILS REVISITED

4.1 General statement

By considering that DRILS is exploring the space of local optima, we would like to improve its ability to discover promising local optima, while avoiding to get stuck. The graybox crossover is very efficient to explore a large proportion of solutions; however, it may be too intensification-oriented when it comes to jump to new regions. By contrast, (iterated) local search can enable to jump more easily to more diverse local optima; however, it may take too many iterations to eventually find high-quality local optima. Besides, the perturbation does not play the same role depending on whether the local search is performed alone, or accompanied with crossover. In particular, the optimal setting of the perturbation strength can be seemingly different depending on the other design choices as shown in our preliminary analysis. Therefore, we investigate the idea of using, *not one, but two customized perturbation strengths*, while carefully *alternating* between local search and crossover in an attempt to infer a more accurate intensification/diversification trade-off. We there-by propose two simple designs, denoted DRILS+ and DRILS++, as depicted respectively in Algorithms 2 and 3.

4.2 Description of DRILS+ and DRILS++

The differences with DRILS are highlighted in gray in the templates of DRILS+ and DRILS++. Both variants use *two* strength values for the perturbation, namely, α and β. The main difference between the two variants is *at what stage* the β-perturbation is performed.

4.2.1 DRILS+. As depicted in Algorithm 2, DRILS+ applies a second perturbation with strength β if the condition of line 7 holds. In fact,

Algorithm 2: DRILS+: a retroactive escape perturbation

```
1  x ← initial solution;
2  p1 ← EfficientHammingBallHC(x);
3  while not stopping condition do
4      x ← Perturbation(p1, α);
5      p2 ← EfficientHammingBallHC(x);
6      p ← PartitionCrossover(p1, p2);
7      if p1 = p or p2 = p then
8          x ← Perturbation(p1, β);
9          p1 ← EfficientHammingBallHC(x);
10     else
11         p1 ← EfficientHammingBallHC(p);
```

Algorithm 3: DRILS++: a proactive escape perturbation

```
1  x ← initial solution;
2  p1 ← EfficientHammingBallHC(x);
3  while not stopping condition do
4      if rand(0, 1) < γ then
5          x ← Perturbation(p1, β);
6          p1 ← EfficientHammingBallHC(x);
7      else
8          x ← Perturbation(p1, α);
9          p2 ← EfficientHammingBallHC(x);
10         p ← PartitionCrossover(p1, p2);
11         if p1 = p or p2 = p then
12             p1 ← p2;
13         else
14             p1 ← EfficientHammingBallHC(p);
```

assume that the crossover produced the same child as its parents. This means that although it has the ability to explore a huge proportion of children, none of them was improving. Assuming that the second parent was generated using an accurate perturbation factor α (maximizing the chances of the crossover to find an improvement), we interpret this as a signal that the search starts to stagnate. Instead of resuming the search from the lastly generated local optimum p_2 (as done in DRILS), DRILS+ resumes the search from a new fresh local optimum obtained by applying a perturbation of strength β on the previous first parent p_1 (lines 8 and 9). The motivation is to move to a local optimum that would not have been found using the perturbation strength α. Let us recall that for the graybox crossover to realize its best potential, the value of α is crucially important, since it should allow the local search to prepare an accurate second parent. Hence, the design of the perturbation in DRILS+ allows us to clearly separate the role of the perturbation in DRILS: (i) the β-perturbation is to be thought with respect to the ability to escape whenever the search is trapped, and (ii) the α-perturbation is with respect to the crossover search ability.

DRILS+ can be viewed as an algorithm where the β-perturbation for promoting diversity is performed in retroactive manner, i.e., the β-perturbation is applied only after detecting that the search is not improving anymore. Another option is to *not* wait for such an event to happen, and to proceed in a proactive manner so as to minimize the probability of being trapped. This is precisely the purpose of our second variant, denoted DRILS++.

4.2.2 DRILS++. As depicted in Algorithm 3, we manage to alternate between performing a pure local search using a β-perturbation, and crossover using an a priori different α-perturbation. More precisely, with probability γ, the first parent p_1 is shaken using the β-perturbation, and it is systematically replaced by the local optima obtained by the subsequent local search. With probability $1-\gamma$, we proceed exactly as in DRILS. From our preliminary study, ILS was found to be worst than DRILS, but it showed some potential in escaping local optima with a different range of the perturbation strength. Hence, the design of DRILS++ aims at promoting diversity by considering a dedicated β-perturbation as in DRILS+. However, it feeds more frequently (with probability γ) the crossover with fresh first parents to explore.

4.3 Experimental set-up

We experiment the same NKQ instances as in our preliminary analysis. We vary α and β in the range $\{0.1, 0.25, 0.5, 1, 2.5, 5, 10, 15\} \cdot 10^{-2}$. For DRILS++, we experiment three values of $\gamma \in \{0.25, 0.5, 0.75\}$. Notice that DRILS++ with the extreme values of $\gamma = 0$ and $\gamma = 1$ degenerates to DRILS and ILS respectively. Overall, we consider $5_{(k)} \times 4_{(n)} \times 8_{(\alpha)} \times 8_{(\beta)} \times (1_{(\text{DRILS+})} + 3_{(\gamma,\text{DRILS++})}) = 5120$ additional configurations requiring 533.3 CPU days to run.

For the sake of our analysis, we compute the values of α, β and γ providing the best average relative fitness deviation (rfd) as defined previously. This is performed on a per-instance basis in order to conduct a fine-grained analysis of the performance of each algorithm when optimally configured. Additionally, we compute the perturbation providing the best average rfd-values with respect to the whole set of instances, as well as, when splitting them into different groups according to k or n. In this scenario, the (best) combination of α and β can be different from the per-instance best setting, and so is the underlying performance. This is intended to provide a more coarse-grained analysis of the impact of perturbation. For each setting, we use a Wilcoxon test at significance level 0.05 for assessing statistical differences.

We first start by analyzing the relative performance of the algorithms when configured optimally. This is followed by a detailed analysis of the impact of the perturbation strength configuration.

4.4 Performance analysis

In Table 1, we report the ranks of the different algorithms obtained when using the best perturbation values. The rank of an algorithm is computed as the number of other algorithms that are significantly better than the considered algorithm. The best possible rank is hence 0. Besides, we indicate in Table 1 which variants are significantly better (▲), worst (▽), or similar (≈) to DRILS.

4.4.1 Overall performance. Over all the instances (bottom-right corner of Table 1), DRILS+ and DRILS++ are both ranked first, with DRILS+ being significantly better than DRILS. This provides a first

Table 1: Ranks and (best) average rfd-values (between the braces). The rfd-values are rounded to two decimals for the sake of presentation. Rank 0 in bold corresponds to algorithms that are *not* significantly outperformed by any other algorithm, and the signs (▲), (▽), and (≈) indicate whether the algorithm under consideration is significantly better, worst or similar to DRILS.

		$k=2$			$k=3$			$k=4$			$k=5$			$k=6$			**per n**		
$n=10^3$	DRILS	**0**	(0.00)		1	(0.11)		1	(0.56)		1	(0.58)		2	(0.33)		1	(0.70)	
	DRILS+	**0**	(0.00)	≈	**0**	(0.00)	▲	**0**	(0.02)	▲	**0**	(0.10)	▲	**0**	(0.06)	▲	**0**	(0.09)	▲
	DRILS++	**0**	(0.00)	≈	1	(0.12)	≈	1	(0.52)	≈	1	(0.50)	≈	1	(0.27)	▲	1	(0.70)	≈
$n=10^4$	DRILS	**0**	(0.00)		1	(0.10)		1	(0.16)		2	(0.13)		**0**	(0.15)		1	(0.40)	
	DRILS+	**0**	(0.00)	≈	**0**	(0.02)	▲	**0**	(0.06)	▲	**0**	(0.06)	▲	**0**	(0.14)	≈	**0**	(0.21)	▲
	DRILS++	**0**	(0.00)	≈	1	(0.09)	≈	1	(0.15)	≈	**0**	(0.09)	▲	**0**	(0.11)	≈	1	(0.40)	≈
$n=10^5$	DRILS	**0**	(0.00)		2	(0.04)		2	(0.21)		2	(0.33)		2	(0.71)		1	(0.28)	
	DRILS+	**0**	(0.00)	≈	1	(0.03)	▲	**0**	(0.05)	▲	**0**	(0.04)	▲	**0**	(0.06)	▲	**0**	(0.07)	▲
	DRILS++	**0**	(0.00)	≈	**0**	(0.01)	▲	1	(0.12)	▲	1	(0.09)	▲	1	(0.17)	▲	1	(0.27)	≈
$n=10^6$	DRILS	**0**	(0.00)		1	(0.02)		1	(0.41)		1	(0.85)		1	(0.98)		1	(0.53)	
	DRILS+	**0**	(0.00)	≈	1	(0.02)	≈	1	(0.41)	≈	1	(0.84)	≈	1	(0.98)	≈	1	(0.53)	≈
	DRILS++	**0**	(0.00)	≈	**0**	(0.00)	▲	**0**	(0.04)	▲	**0**	(0.05)	▲	**0**	(0.04)	▲	**0**	(0.25)	▲
per k	DRILS	**0**	(0.00)		2	(0.07)		1	(0.34)		1	(0.53)		1	(0.65)		1	(0.48)	
	DRILS+	**0**	(0.00)	≈	**0**	(0.02)	▲	**0**	(0.26)	≈	**0**	(0.44)	≈	1	(0.66)	≈	**0**	(0.39)	▲
	DRILS++	**0**	(0.00)	≈	1	(0.06)	▲	**0**	(0.25)	▲	**0**	(0.29)	▲	**0**	(0.32)	▲	**0**	(0.43)	≈

(coarse-grained) empirical evidence on the importance of an accurate design of the perturbation when hybridizing the different graybox techniques. In the following, we analyze more carefully the relative performance as a function of the landscape characteristics.

4.4.2 Impact of the landscape difficulty (k). Having a look at the different ranks for the instances *grouped by k* (i.e., last row of Table 1), DRILS+ is found to be significantly better than DRILS for $k = 3$, whereas it has a comparable performance for large values of $k = 4, 5, 6$. By contrast, DRILS++ is significantly better than DRILS for the large values of $k = 4, 5, 6$. For $k = 2$ (quadratic instances), all algorithms are able to attain an rfd-value of (almost) 0% (notice that we rounded the rfd to two decimals for the sake of presentation). This means that for easy landscapes, the graybox operators are extremely effective; however, they start facing difficulty as the problem ruggedness increases with k. The perturbation enters into play and allows to observe significant differences with DRILS.

Since DRILS+ waits until the search does not improve before performing the β-perturbation, it gives more time to the crossover to intensify the search around the current parents, while providing a reasonable chance to escape. However, for very large and rugged landscapes, this seems to be not sufficient to counter-balance the extreme intensification ability of the crossover, and the search is very likely trapped more easily. By contrast, the proactive design of DRILS++ allows to anticipate such a situation by promoting diversity without necessary waiting for the crossover to intensively explore the lastly discovered parents. Interestingly, the only group of instances for which DRILS+ is significantly out-performed by DRILS++ is for $k = 6$, which is the most difficult instances; and the only instances where DRILS++ is significantly out-performed by DRILS+ is for $k = 3$. This renders the different intensification-diversification trade-offs of these two alternative perturbation schemes.

These observations hold only when the algorithms are configured with the best strength for a *set* of instances *grouped by k*, i.e., a *global* configuration taking into account the heterogeneity of the instances. The situation is clearly different when having a more fine-grained per-instance configuration taking into account the dimension n, which is analyzed next.

4.4.3 Impact of the landscape dimension (n). Looking at the relative performance following the landscape dimension n (e.g., last column of Table 1), DRILS+ is found to be overall more effective than DRILS++ for almost all instances of dimension $n \leq 10^5$ except for $n = 10^5$ and $k = 3$, and $n = 10^4$ and $k = 5, 6$. For the largest one-million variable instances with $k \geq 3$, DRILS++ out-performs DRILS+ significantly. Actually, this can be further appreciated by analyzing the anytime behavior of the different algorithms. Due to space restrictions, this is illustrated in Fig. 2 for a set of representative instances with relatively small and large n and k.

For the smallest instance, DRILS+ is by-far better till the very early stages of the search. As n (or k) increases, DRILS++ is able to reduce the gap with DRILS+ eventually being better. For the most difficult instance ($n = 10^6$, $k = 6$), DRILS++ performs similarly to DRILS and DRILS+ in the early stages; but it becomes superior as the search is advancing. Notice that the profiles of DRILS and DRILS+ are overlapping in Fig. 2 for this instance.

4.4.4 Crossover success probability. To complement our study, we analyze how the ability of the crossover to find improving children is impacted by the perturbation. For this purpose, we did an additional experiment using two instance sets with $n \in \{10^4, 10^5\}$ and $k \in \{4, 5, 6\}$. For each algorithm execution, we record the number of times the crossover was successful in finding an improving child all along the run. We also record the number of times an improving child was it-self improved by the hill-climber performed just after the crossover. We then report the (empirical) success probability of the partition crossover, and the success probability of the subsequent hill-climber (or equivalently the probability that the child is *not* a local optimum). This is shown in Fig. 3 with respect to the (per-instance) best configuration of each algorithm, and aggregating the probability values over the 10 runs and the k-values.

We can see that the partition crossover success probability is the highest for DRILS++. It is almost 1 for $n = 10^5$, whereas it is around 0.8 for DRILS and DRILS+. This can only be attributed to the

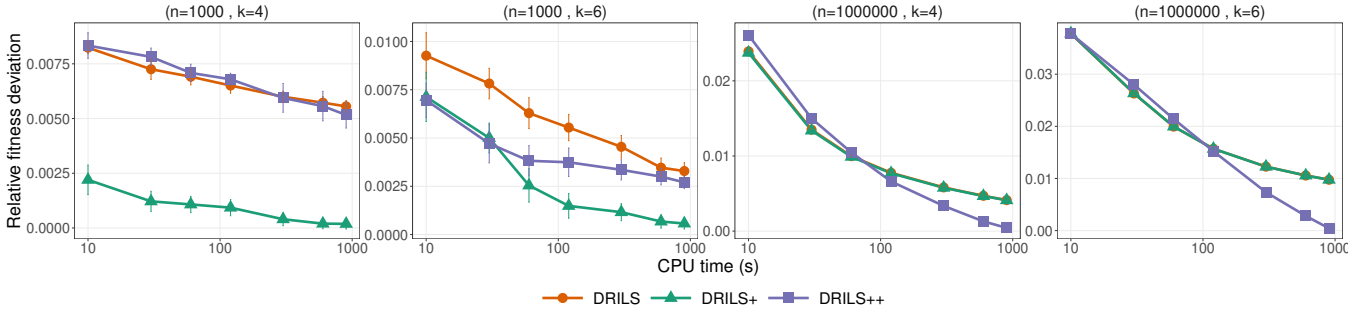


Figure 2: Example of the runtime behavior on four NKQ landscapes with increasing n and k values.

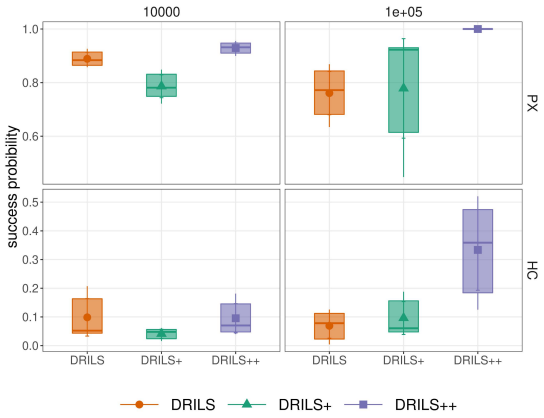


Figure 3: Success probability of the partition crossover (PX, top) and the following hill-climber (HC, bottom) for $n \in \{10^4, 10^5\}$ and $k \in \{4, 5, 6\}$.

design of the β-perturbation. Since $\gamma > 0$, DRILS++ performs overall less crossovers than DRILS and DRILS+. Interestingly, this does not decrease the chance to find improving children. On the contrary, the β-perturbation allows DRILS++ to move to fresh local optima for which recombination is found to be more likely successful. Besides, the probability that the improving children found with DRILS++ are not local optima increases. This can be explained by the ability of DRILS++ to visit diversified search regions where it is more likely to find improvements. Notice that this does not necessary mean that DRILS++ is systematically improving the best solution, since the success probability is computed with respect to the *current* first parent. However, this shows that the diversity introduced by the β-perturbation can be beneficial for the crossover success probability, and subsequently for the search process as a whole.

4.5 Analysis of the perturbation strength

In this section, we analyze the configuration of the perturbation *strength*. In Table 2, we report the values of α, β, and γ, that provide the best rfd-value for every algorithm. We report the values computed on a per-instance basis, as well as, with respect to the different sets of instances grouped by n or k.

4.5.1 Impact of α. Firstly, we can see that the optimal setting of α follow the same trend across the different algorithms and instances. For small values of $k = 2, 3$, the best perturbation strength α falls in the highest range of experimented values ($\alpha = 15\%, 10\%$). For the larger values of $k = 4, 5, 6$, the best α-value decreases gradually to 2.5%. This is actually consistent for all algorithms, and in line with our preliminary analysis. Thus, we can conclude that the α-perturbation is definitively to be set so that the *second* parent is accurate to recombine with the first one. In other words, the best setting of α should be guided by the ability of feeding the crossover with accurate recombination parents as a function of k. Let us recall that in the original design of DRILS, the α-perturbation plays two roles: generating a second parent for the crossover *and* updating the next first parent in case the crossover does not improve. By contrast, DRILS+ and DRILS++ make a clear difference between these two roles by introducing the second β-perturbation.

4.5.2 Impact of β. Interestingly, looking at the values of β, we can report two general tendencies, which are further supported by Fig. 4 providing a more detailed view of the interaction between α and β. On the one hand, the best β-values is typically smaller than α for both DRILS+ and DRILS++ (e.g., see how the knee in Fig. 4 is shifted to the left as k increases). However, as the instance difficulty increases, the gap between the best α and β decreases. This clearly indicates that for optimal performances, the perturbation needed to feed the crossover, should be different from the perturbation needed for escaping local optima, which is precisely what is achieved by DRILS+ and DRILS++ as supported by the previous performance analysis. Besides, the fact that the value of β is typically smaller than α might seem surprising since β is to be viewed as diversification-oriented. Nevertheless, this is to be considered with respect to the non-elitist acceptance criteria which complements the β-perturbation. On the other hand, we found that DRILS+ requires a larger value of β in comparison to DRILS++ especially for the largest values of k. This is clearly due to the different intensification-diversification trade-offs implied by these two designs. In fact, it is more likely that the crossover has already reached a high-quality first parent when the β-perturbation is activated by DRILS+. Thus, the perturbation strength β is to be set relatively larger than for DRILS++, since DRILS++ applies continuously the β-perturbation with probability γ. With respect to DRILS++, the value of β should be relatively lower to avoid having a too diversified search, and give a chance to the crossover to work at its best.

4.5.3 Impact of γ. From Table 2, the best performing values of γ are mostly 0.25 and 0.5. This means that performing the β-perturbation

Table 2: The values (in percentage) of α, β and γ providing the best average relative fitness derivation.

		$k=2$			$k=3$			$k=4$			$k=5$			$k=6$			per n		
		α	β	γ	α	β	γ	α	β	γ	α	β	γ	α	β	γ	α	β	γ
$n=10^3$	DRILS	15			15			5			0.1			0.2			0.2		
	DRILS+	10	0.1		10	0.1		15	1		2.5	1		2.5	1		5	1	
	DRILS++	15	0.2	25	15	2.5	25	5	0.5	25	0.1	0.1	75	0.2	0.1	75	0.2	0.1	25
$n=10^4$	DRILS	15			15			5			2.5			2.5			2.5		
	DRILS+	10	0.1		15	2.5		5	2.5		5	2.5		2.5	2.5		5	1	
	DRILS++	15	0.1	50	15	0.1	50	5	0.2	25	2.5	0.2	25	2.5	0.1	50	2.5	0.2	25
$n=10^5$	DRILS	15			15			2.5			5			2.5			2.5		
	DRILS+	15	0.2		5	15		5	10		2.5	5		2.5	5		2.5	5	
	DRILS++	15	0.2	25	15	0.5	25	5	0.5	75	2.5	1	50	2.5	1	50	5	1	25
$n=10^6$	DRILS	15			15			5			5			2.5			2.5		
	DRILS+	15	1		15	15		5	10		5	15		2.5	1		2.5	0.5	
	DRILS++	15	0.1	25	15	1	25	5	2.5	25	2.5	1	50	2.5	1	50	2.5	1	25
per k	DRILS	15			15			5			2.5			2.5			2.5		
	DRILS+	15	1		15	2.5		5	2.5		5	2.5		2.5	2.5		2.5	1	
	DRILS++	15	0.2	25	15	1	25	5	1	25	2.5	1	50	2.5	1	50	2.5	1	25

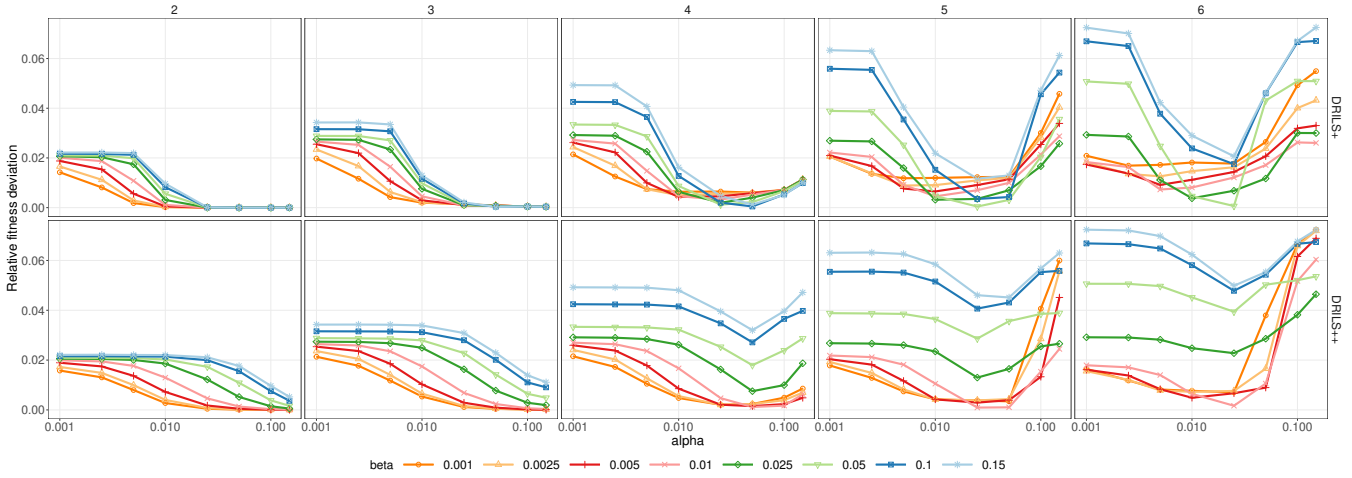


Figure 4: rfd-values for DRILS+ (top) and DRILS++ (bottom) as a function of α and β. Left to right: $k = 2, 3, 4, 5$ and 6. $n = 10^5$, $\gamma = 0.5$.

half or quarter of the time, is a fairly good choice. Since the best β-perturbation was found to be relatively small, such a setting allows in fact to visit new local optima leading the crossover to find high-quality improvements.

5 CONCLUSION AND PERSPECTIVES

In this paper, we revisited the design of the perturbation mechanism in the state-of-the-art DRILS framework, ending up with enhanced hybridization schemes exposing new intensification-diversification trade-offs. Beyond being able to achieve significant improvements, our work provides evidence on the importance of perturbation in graybox optimization techniques. In particular, the impressive ability of the partition crossover to explore a large number of solutions, should be balanced with carefully designed escape mechanisms taking into account the dimension and the ruggedness of the tackled landscapes. Our work demonstrates that the perturbation can in fact play different roles, influencing the ability of the graybox crossover to keep improving, and allowing to visit unexplored promising regions. Besides, a number of questions are left open.

The factorial experimental design adopted in this work allows us to gain a more fundamental understanding of the complex interactions between the best setting of the perturbation parameters and the landscape characteristics. However, the tuning of the different perturbation parameters may be an issue in practice. Deriving adaptive and landscape-informed strategies constitutes a challenging research direction that can, not only alleviate the burden of a manual calibration, but lead to improved performances.

Another research perspective would be to study other algorithmic aspects in DRILS. For instance, we have shown that an elite variant of DRILS is not a good choice. This relates to the acceptance criteria of standard ILS, which is known to be complementary to the perturbation. In DRILS, we have to consider, not one, but two parents, for recombination at each iteration. One interesting direction would be to investigate the smart combination of the perturbation with the search history, in order to design informed mechanisms for selecting which local optima are worth to be shaken and recombined at each iteration.

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